

Advanced Econometrics – Exam

June 20, 2022

Exercise 1 (30%)

In this example, we have (synthetic) data¹ on whether workers complain to managers at fast-food restaurants. The covariates are **age** (in years of the worker), **grade** (years of schooling completed by the worker), **south** (equal to 1 if the restaurant is located in the South), **tenure** (the number of years spent on the job by the worker), **gender** (of the worker, 1 for women), **race** (of the worker), **income** (in thousands of dollars by the restaurant), **genderm** (gender of the manager, 1 for women), **burger** (equal to 1 if the restaurant specializes in hamburgers), and **chicken** (equal to 1 if the restaurant specializes in chicken).²

Call:

```
glm(formula = complain ~ age + grade + south + tenure + gender +
     income, family = binomial(link = "probit"), data = chicken)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.4023853	0.5083557	0.792	0.42863
age	-0.0336174	0.0118943	-2.826	0.00471 **
grade	0.0331282	0.0243458	1.361	0.17360
south	0.0776746	0.0547301	1.419	0.15583
tenure	-0.0029834	0.0899963	-0.033	0.97355
gender	0.2975866	0.0547633	5.434	5.51e-08 ***
income	-0.0020855	0.0006771	-3.080	0.00207 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2804.7 on 2762 degrees of freedom

Residual deviance: 2752.6 on 2756 degrees of freedom

AIC: 2766.6

Call:

```
probitmfx(formula = complain ~ age + grade + south + tenure +
           gender + income, data = chicken, atmean = TRUE)
```

Marginal Effects:

	dF/dx	Std. Err.	z	P> z
age	-0.00942873	0.00333170	-2.8300	0.004655 **
grade	0.00929152	0.00682659	1.3611	0.173489
south	0.02174921	0.01529361	1.4221	0.154994
tenure	-0.00083676	0.02524139	-0.0332	0.973555
gender	0.08348932	0.01529146	5.4599	4.765e-08 ***
income	-0.00058492	0.00018957	-3.0855	0.002032 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

dF/dx is for discrete change for the following variables:

[1] "south" "gender"

¹The data set is available at <https://www.stata-press.com/data/r17/chicken.dta>.²The exercise is based on Stata Reference Manual (xtprobit function).

```
> null_probit = glm(complain~1, data=chicken, family=binomial(link="probit"))
> lrtest(probit1, null_probit)
Likelihood ratio test
```

Model 1: complain ~ age + grade + south + tenure + gender + income

Model 2: complain ~ 1

```
#Df  LogLik Df Chisq Pr(>Chisq)
1    7 -1376.3
2    1 -1402.3 -6 52.11  1.773e-09 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> gof.results = gof(probit1)
```

```
> gof.results$gof
```

	test	stat	val	df	pVal
1:	HL	chiSq	4.9805893	8	0.7596487
2:	mHL	F	0.7304362	9	0.6813310
3:	OsRo	Z	1.2660267	NA	0.2055035
4:	SstPgeq0.5	Z	NaN	NA	NaN
5:	SstPl0.5	Z	0.3283324	NA	0.7426603
6:	SstBoth	chiSq	NaN	2	NaN
7:	SllPgeq0.5	chiSq	0.0000000	1	1.0000000
8:	SllPl0.5	chiSq	0.1081492	1	0.7422611
9:	SllBoth	chiSq	0.1081492	2	0.9473614

Use $\alpha = 5\%$ significance level.

- (5%) Interpret parameters' estimates.
- (5%) Interpret marginal effects.
- (5%) Are all variables jointly statistically significant? Explain.
- (7%) Let's focus on variable **tenure**. Is it endogenous? Explain.
- (8%) Interpret the diagnostic tests' results.

Exercise 2 (30%)

University administrators want to know the relationship between high school grade point average (GPA) and students' performance in college. `gpa.dta` contains fictional data on a cohort of 4,000 college students.³ College GPA (`gpa2`) and high school GPA (`hsgpa`) are measured on a continuous scale between zero and four. The outcome of interest is the student's college GPA. But, for reasons of confidentiality, GPAs below 2.0 are reported as 2.0. In other words, the outcome is censored on the left.

We believe that GPA is also a function of the logarithm of income of the student's parents (`pincome`) and whether or not the student participated in a study-skills program while in college (`program`).⁴

The output below presents a tobit model estimates and the table presents marginal effects for this model.

Call:

```
censReg(formula = gpa2 ~ hsgpa + pincome + program, left = 2,
        right = Inf, data = gpa)
```

Observations:

Total	Left-censored	Uncensored	Right-censored
4000	1206	2794	0

Coefficients:

	Estimate	Std. error	t value	Pr(> t)
(Intercept)	-0.890258	0.047848	-18.61	<2e-16 ***
hsgpa	0.658631	0.012870	51.18	<2e-16 ***
pincome	0.315930	0.007457	42.37	<2e-16 ***
program	0.555442	0.014747	37.66	<2e-16 ***
logSigma	-0.910997	0.013606	-66.95	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Newton-Raphson maximisation, 7 iterations

Return code 1: gradient close to zero

Log-likelihood: -2015.126 on 5 Df

```
> x = rbind(1, mean(gpa$hsgpa), mean(gpa$pincome), mean(gpa$program))
> me = tobit_marginal_effects(tobit1, x, dummies_indices=c())
```

Marginal effects of the tobit model

	y*	E(y x)	E(y x,y>0)	Pr(y>0 x)	at X=
hsgpa	0.6586311	0.6586311	0.6586311	8.750838e-09	2.842114
pincome	0.3159297	0.3159297	0.3159297	4.197569e-09	3.680588
program	0.5554416	0.5554416	0.5554416	7.379821e-09	0.498500

³The data set is available at <https://www.stata-press.com/data/r16/gpa.dta>.

⁴The exercise is based on Stata Reference Manual (`tobit` function).

Use $\alpha = 5\%$ significance level.

- a) (5%) Why shouldn't we use a simple regression model for the problem? Explain.
- b) (7%) Is this a corner solution or a censoring variable case? Explain.
- c) (5%) What is the interval of values of the dependent variable? How many observations did take values on the edge of the interval?
- d) (5%) Which variables are individually significant? Explain.
- e) (8%) What is the effect of participation in a study-skills program while in college? Use appropriate marginal effect.

Exercise 3 (40%)

A certain professor wanted to prepare an easy exam question for the students he likes (really!). After a moment of thought, he decided to use a panel data model. The dependent variable was the number of mobile phones per country in a given year, while the regressor was the number of fixed phones (landlines) per country in a given year. The data span was from 1960 to 2011.

Call:

```
plm(formula = Mobile ~ FixedPhones, data = phones,
     model = "within", index = c("Country or Area", "Year"))
Unbalanced Panel: n = 181, T = 0-37, N = 5800
```

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
FixedPhones	2.391561	0.051165	46.743	< 0.00000000000000022 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 188920000000000000

Residual Sum of Squares: 136020000000000000

R-Squared: 0.9994965

Adj. R-Squared: 0.9970201

F-statistic: 2184.87 on 1 and 5618 DF, p-value: < 0.000000000000000222

Use $\alpha = 5\%$ significance level.

- (5%) Interpret the parameter next to **FixedPhones**.
- (5%) How many countries were included in the data set? Explain.
- (5%) Was it the fixed effects or the random effects estimator? Explain.
- (5%) The output below presents a stationarity analysis of the variable **Mobile** for one of the countries in the data set (Poland). What is the conclusion? Explain.
- (10%) The next output below presents a stationarity analysis of the variable **Mobile** for a different country in the data set (Sweden). However, the lecturer decided to present the results of two tests. What is the conclusion? What should be done?
- (5%) Is the relationship between the number of mobile phones and fixed phones real or spurious? Explain.
- (5%) Is there a way to improve the presented panel model? Explain.

The output for the number of mobile phones for Poland.

```
> testdf2(variable = Poland$Mobile, # vector tested
+         test.type = "c", # test type
+         max.augmentations = 3, # maximum number of augmentations added
+         max.order=3) # maximum number of analysed autocorrelation
  augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 0.9577101  0.99             11906          33.2867158237
2             1 1.1022916  0.99             11905           0.0451182404
3             2 1.2029169  0.99             11904           0.0031990020
4             3 1.2413343  0.99             11903           0.0007007256
  bgodfrey.statistic.2 bgodfrey.statistic.3      p_bg.p.value.1      p_bg.p.value.2
1          49.36518303          51.70576242 0.0000000007952402 0.00000000001907597
2          16.74534552          18.75293892 0.831786758303942 0.00023109706022170
3           0.02579603           3.18590948 0.954895921646715 0.98718480814492915
4           0.01667437           0.08902341 0.978881486915712 0.99169747171767997
      p_bg.p.value.3
1 0.000000000003460303
2 0.00030751406160314
3 0.36384015909338202
4 0.99312130594989223
```

The output for the number of mobile phones for Sweden.

```
#####
# Phillips-Perron Unit Root Test #
#####
```

Test regression with intercept

Value of test-statistic, type: Z-tau is: -116.0626

```
      aux. Z statistics
Z-tau-mu      2.2604
```

Critical values for Z statistics:

```
      1pct    5pct    10pct
critical values -3.434004 -2.86233 -2.567221
```

```
#####
# KPSS Unit Root Test #
#####
```

Test is of type: mu with 13 lags.

Value of test-statistic is: 74.8118

Critical value for a significance level of:

```
      10pct  5pct 2.5pct  1pct
critical values 0.347 0.463 0.574 0.739
```


